CACNA1I-Related Neurodevelopmental Disorder — Comprehensive Research Report

Disease: CACNA1I-Related Neurodevelopmental Disorder **MONDO ID:** MONDO:0859313 **Causal gene: CACNA1I** (encoding the CaV3.3 low-voltage-activated T-type calcium channel, HGNC:1396) **Synonym:** Neurodevelopmental disorder with speech impairment and with or without seizures **Category:** Mendelian (autosomal dominant)

> **Causal-gene statement (per template):** This report concerns the disease caused by **CACNA1I**, which encodes the pore-forming α1 subunit of the **CaV3.3** (α1I) low-voltage-activated T-type calcium channel. The symbol CACNA1I is easily confused with other voltage-gated calcium channel genes (CACNA1A, CACNA1C, CACNA1G, CACNA1H); all content below is specific to CACNA1I / CaV3.3.

Summary

CACNA1I-related neurodevelopmental disorder is a rare, autosomal-dominant channelopathy caused by heterozygous — usually *de novo* — missense variants in *CACNA1I*, the gene encoding the CaV3.3 low-voltage-activated (T-type) calcium channel. The disorder is defined clinically by global developmental delay/intellectual disability, prominent speech impairment, hypotonia, and variable epilepsy. It was first established as a disease gene by El Ghaleb and colleagues in 2021 (PMID: 33704440), who identified four heterozygous missense variants in patients with variable neurodevelopmental phenotypes, and was consolidated in 2025 with a further allelic series (PMID: 40825030).

Mechanistically, the majority of pathogenic variants cluster at the cytoplasmic ends of the pore-lining S5/S6 segments — the channel activation gate — and predominantly cause **gain of channel function**: slowed activation/inactivation/deactivation kinetics and hyperpolarizing (leftward) shifts in the voltage dependence of gating, which together increase neuronal excitability. An allelic series demonstrates the effect is residue- and substitution-specific: at

codon 398, A398E is gain-of-function while A398V is loss-of-function, showing opposite biophysical outcomes can arise at a single position. CaV3.3 is highly expressed in the GABAergic neurons of the **thalamic reticular nucleus (TRN)**, where its pacemaking burst-firing activity generates **sleep spindles** during NREM sleep — a role confirmed in CaV3.3-knockout mice. Disruption of this thalamocortical rhythmogenesis provides a coherent mechanistic bridge from the molecular lesion to the speech-predominant neurodevelopmental phenotype.

There are no disease-specific therapies. Management is supportive (speech, occupational, and behavioral therapy; antiseizure medication for seizures). Because the disorder is predominantly a gain-of-function T-type channelopathy, **T-type calcium channel blockers** (ethosuximide, zonisamide, and investigational selective blockers such as apinocaltamide/ACT-709478) offer a biologically rational but as-yet-unvalidated precision-therapy avenue, supported by evidence from the paralogous CACNA1G and CACNA1H T-type channelopathies. Importantly, this Mendelian disorder is mechanistically distinct from the separate, polygenic association of common *CACNA1I* variation with schizophrenia.

Key Findings

Finding 1 — CACNA1I (CaV3.3) is the causal gene for an autosomal-dominant NDD

Heterozygous missense variants in *CACNA1I* cause a neurodevelopmental disorder with variable severity. The seminal study identified **four heterozygous missense variants** clustering at the cytoplasmic ends of the pore-lining S5/S6 segments: p.Ile860Met and p.Ile860Asn in IIS6, p.Ile1306Thr in IIIS5, and p.Met1425Ile in IIIS6 (PMID: 33704440). The follow-up study added **three further activation-gate-modifying variants** (p.Met1425Val, p.Ala398Val, p.Ala398Glu) across four unrelated patients, establishing a formal allelic series and confirming *CACNA1I* as "a disease gene for neurodevelopmental disease with or without epilepsy" (PMID: 40825030).

> *"we examined four heterozygous missense variants in CACNA1I, encoding the Cav3.3 channel, in patients with variable neurodevelopmental phenotypes"* — PMID: 33704440

> *"we established CACNA1I, the gene coding for CaV3.3, as a disease gene for neurodevelopmental disease with or without epilepsy"* — PMID: 40825030

Evidence type: Human clinical + in vitro functional. **HGNC:** CACNA1I; **chromosome:** 22q13.1.

Finding 2 — Disease variants predominantly cause gain of CaV3.3 channel function via activation-gate modification

In HEK293T patch-clamp electrophysiology, pathogenic variants **slow the kinetics of activation, inactivation, and deactivation** and **hyperpolarize (left-shift) the voltage-dependence of activation and inactivation**, net-increasing calcium influx and neuronal excitability (PMID: 33704440; PMID: 40825030). The allelic series demonstrated that outcomes are substitution-specific: **A398E = gain of function**, whereas **A398V = loss of function** (decreased current density, accelerated gating); M1425V and M1425I both cause gain of function.

> *"The mutations resulted in slowed kinetics of current activation, inactivation, and deactivation, and in hyperpolarizing shifts of the voltage-dependence of activation and inactivation"* — PMID: 33704440
>
> *"the two substitutions of A398 show opposite effects on channel function. While substitution A398E leads to a gain of channel function, A398V results in decreased current density"* — PMID: 40825030

Functional consequence: predominantly gain of function; loss of function at some residues. **Evidence type:** in vitro functional (heterologous expression).

Finding 3 — CaV3.3 pacemaking in TRN GABAergic neurons links the channelopathy to sleep spindles

CaV3.3 is expressed in **GABAergic neurons of the thalamic reticular nucleus (TRN)**, where pacemaking activity controls **sleep-spindle rhythmogenesis during NREM sleep** (PMID: 40825030). The schizophrenia-associated R1346H variant reduces synaptic CaV3.3 and consequently sleep spindles (PMID: 36865984). More broadly, CaV3.3 governs low-threshold calcium spikes, burst firing, and rhythmic oscillations involved in sensory processing and sleep (PMID: 33704440).

> *"CaV3.3 is expressed in GABAergic neurons of the thalamic reticular nucleus (TRN), where its pacemaking activity controls sleep spindle rhythmogenesis during the non-rapid eye movement (NREM) phase of natural sleep"* — PMID: 40825030
>
> *"the R1346H variant in the CACNA1I gene coding for the Cav 3.3 protein results in a synaptic reduction of Cav3.3 voltage gated calcium channels and, consequently, sleep spindles"* — PMID: 36865984

Cell type (CL): GABAergic interneuron (CL:0000617); thalamic reticular nucleus neuron. **Anatomy (UBERON):** thalamic reticular nucleus (UBERON:0001903).

Finding 4 — Core clinical phenotype: developmental delay/ID, speech impairment, hypotonia, variable epilepsy

Reported patients display **global developmental delay, variable cognitive impairment, hypotonia, and epilepsy/seizures**, with severity ranging from mild (familial p.Ile860Met) to severe (de novo p.Ile860Asn, with developmental delay and seizures) (PMID: 33704440). The disorder's designation — *neurodevelopmental disorder with speech impairment and with or without seizures* — reflects the prominence of speech/language deficits. An autism spectrum disorder (ASD) case with profound speech impairment, stereotypies, sensory dysregulation, and global delay carried a *CACNA1I* VUS (p.Ala2010Ser) (PMID: 41798403).

> *"The de novo p.(Ile860Asn) variant, changing the same amino acid residue, was detected in a patient with severe developmental delay and seizures"* — PMID: 33704440
>
> *"In two additional individuals with global developmental delay, hypotonia, and epilepsy"* — PMID: 33704440

Suggested HPO terms: Global developmental delay (HP:0001263); Intellectual disability (HP:0001249); Delayed speech and language development / Speech impairment (HP:0000750); Muscular hypotonia (HP:0001252); Seizure (HP:0001250); Autistic behavior (HP:0000729).

Finding 5 — Dual genetic relationship to disease: rare Mendelian NDD vs. common/rare schizophrenia variants

CACNA1I has **two distinct disease relationships**. Large-scale GWAS implicate common *CACNA1I* SNPs in schizophrenia susceptibility. Functional analysis of 57 rare/common missense variants from a Swedish cohort of >10,000 found that **reduced CaV3.3 current density dominated control-derived rare alleles**, whereas schizophrenia-patient alleles encoded channels with **altered voltage responses**; pore variants alter current density while voltage-sensor variants alter voltage dependence (PMID: 34919654). Ultra-rare disruptive VGCC variants (including *CACNA1I*) were also found in Japanese schizophrenia/ASD samples (PMID: 35220405).

> *"reduced CaV3.3 channel current density was the dominant phenotype associated with rare CACNA1I coding alleles derived from control subjects, whereas rare CACNA1I alleles from schizophrenia patients encoded CaV3.3 channels with altered responses to voltages" — PMID: 34919654*

Interpretation: The Mendelian NDD (gain-of-function, large effect size) is mechanistically and genetically distinct from the polygenic schizophrenia association. Care must be taken not to conflate them.

Finding 6 — CaV3.3 knockout mice confirm its dominant role in TRN rhythmic bursting and sleep-spindle generation

In **CaV3.3$^{-/-}$ mice**, rhythmic burst discharges (RBDs) in TRN neurons are abolished while tonic firing increases; these mice (and CaV3.2/CaV3.3 double-KO) show increased susceptibility to drug-induced spike-and-wave discharges and absence seizures (PMID: 25071191). **CaV3.2/CaV3.3 double-KO mice** have fragmented NREM sleep with suppressed sigma (10–15 Hz spindle) band power and increased delta; CaV3.3 dominates nRt rhythmogenesis and mediates a substantial fraction of spindle power (PMID: 26612388).

> *"the RBDs were completely abolished, whereas tonic firing was significantly increased, in TRN neurons from mice in which the gene for the T-type Ca(2+) channel, CaV3.3, was deleted (CaV3.3(-/-))" — PMID: 25071191*

> *"The CaV3.3 subtype dominates nRt rhythmic bursting and mediates a substantial fraction of spindle power in the NREM sleep EEG"* — PMID: 26612388

Note on directionality: The mouse models are *loss-of-function* (knockout), whereas human disease is predominantly *gain-of-function*. They converge on the same circuit (TRN pacemaking → thalamocortical rhythmogenesis) but perturb it in opposite directions, consistent with the paradoxical seizure susceptibility observed in both deficient and overactive states.

Finding 7 — Emerging phenotype expansion includes microcephaly and a Dandy-Walker variant, but the core disorder remains non-structural

A 16-month-old with heterozygous *CACNA1I* NM_021096.4:c.4891T>A (p.Phe1631Ile) presented with **congenital microcephaly, frequent seizures, severe global developmental delay, and a Dandy-Walker variant** (partial cerebellar vermis agenesis) on MRI; trio-WES identified this variant in a patient described as having "multiple genetic etiologies," with chromosome-11 regions of homozygosity (PMID: 41791021). Across the better-characterized cohorts, features are developmental delay/ID, speech impairment, hypotonia, and variable seizures **without consistent structural brain malformation** — so the structural findings should be interpreted cautiously given the reported multiple etiologies.

> *"Trio-based whole-exome sequencing identified a heterozygous variant (NM_021096.4:c.4891T>A/p.Phe1631Ile)"* — PMID: 41791021

Finding 8 — T-type channel blockers provide a genotype-guided precision-therapy rationale

The triple T-type blocker **ACT-709478 (apinocaltamide)** suppressed spike-and-wave discharges in WAG/Rij and GAERS absence-epilepsy rat models with efficacy equal/superior to first-line monotherapy and synergy with valproate (PMID: 40165594). Ethosuximide and zonisamide also block T-type current. The paralogous T-type channelopathy — de novo gain-of-function *CACNA1G* (CaV3.1) variants — causes childhood NDD with cerebellar atrophy, severe motor/cognitive impairment, and variable microcephaly and epilepsy, reinforcing both the disease logic and the plausibility of microcephaly in *CACNA1I* disease (PMID: 29878067).

> *"ACT-709478 suppressed SWDs in WAG/Rij rats and GAERS with efficacy equivalent to or superior to that of first-line monotherapy"* — PMID: 40165594
>
> *"We identified four patients with de novo CACNA1G mutations. They all display severe motor and cognitive impairment, cerebellar atrophy as well as variable features such as facial dysmorphisms, digital anomalies, microcephaly and epilepsy"* — PMID: 29878067

Suggested NCIT term: Calcium Channel Blocker (NCIT:C593). **CHEBI:** ethosuximide (CHEBI:4891); zonisamide (CHEBI:10127).

Section-by-Section Report

1. Disease Information

CACNA1I-related neurodevelopmental disorder (MONDO:0859313) is a rare monogenic disorder in which heterozygous missense variants in *CACNA1I* — the gene encoding the **CaV3.3 low-voltage-activated T-type calcium channel** — disrupt thalamocortical rhythmogenesis and produce a neurodevelopmental phenotype dominated by developmental delay, speech impairment, hypotonia, and variable seizures.

Identifier	Value
MONDO	MONDO:0859313
Gene	CACNA1I (HGNC:1396; NCBI Gene 8911)
OMIM	Not formally recorded for the disease; *CACNA1I* gene OMIM *608230
Synonym	Neurodevelopmental disorder with speech impairment and with or without seizures
Inheritance	Autosomal dominant (usually de novo)

Information is derived from **aggregated disease-level resources and individual published case reports / functional studies**, not from EHR-scale cohorts — the total number of reported patients is small (fewer than ~15 well-characterized individuals).

2. Etiology

Causal factor: genetic — heterozygous (predominantly de novo) missense variants in *CACNA1I*. There is no known infectious, environmental, or toxic cause.

- **Genetic risk factors:** the causal variants themselves (activation-gate missense changes; see Section 4). No established modifier genes.
- **Environmental risk factors:** none identified. The disorder is Mendelian with large effect size.
- **Protective factors:** none identified.
- **Gene–environment interactions:** none established. (By contrast, in the *schizophrenia* context, developmental oxidative stress interacting with parvalbumin-interneuron T-type channels has been proposed — PMID: 35079122 — but this pertains to the polygenic association, not the Mendelian NDD.)

3. Phenotypes

Phenotype	Type	HPO term	Onset	Severity	Frequency
Global developmental delay	Clinical sign	HP:0001263	Infancy/ childhood	Mild–severe (variable)	Very frequent
Intellectual disability	Clinical sign	HP:0001249	Childhood	Variable	Frequent
Speech/language impairment	Clinical sign	HP:0000750	Early childhood	Prominent; can be profound	Characteristic (defining)
Muscular hypotonia	Clinical sign	HP:0001252	Neonatal/ infancy	Variable	Frequent
Seizures/epilepsy	Clinical sign	HP:0001250	Variable	Variable (with or without)	Subset
Autistic behavior	Behavioral	HP:0000729	Early childhood	Variable	Subset
Microcephaly (rare)	Physical	HP:0000252	Congenital	Severe	Rare/expansion

Progression: The neurodevelopmental deficits are generally **stable (non-degenerative)** developmental impairments; seizures, when present, are episodic. **Quality-of-life impact** is driven chiefly by the speech/communication deficit and intellectual disability, which affect education, independence, and social functioning; patients typically require multidisciplinary intervention (PMID: 41798403).

4. Genetic / Molecular Information

Causal gene: *CACNA1I* (HGNC:1396; NCBI Gene 8911; chromosome 22q13.1), encoding the pore-forming α1 subunit of CaV3.3.

Pathogenic variants (reported allelic series):

Variant (protein)	Domain	Functional effect	Inheritance	Source
p.Ile860Met (IIS6)	Activation gate	Gain of function	Familial (milder)	P 33704440
p.Ile860Asn (IIS6)	Activation gate	Gain of function	De novo (severe + seizures)	P 33704440
p.Ile1306Thr (IIIS5)	Pore-lining	Gain of function	—	P 33704440
p.Met1425Ile (IIIS6)	Activation gate	Gain of function	—	P 33704440
p.Met1425Val (IIIS6)	Activation gate	Gain of function	—	P 40825030
p.Ala398Glu	Gate-modifying	Gain of function	—	P 40825030
p.Ala398Val	Gate-modifying	Loss of function	—	P 40825030
p.Phe1631Ile	—	NDD + microcephaly/DWV (multiple etiologies)	De novo	P 41791021
p.Ala2010Ser (VUS)	—	VUS; ASD phenotype	Possibly de novo	P 41798403

- **Variant classification:** pathogenic/likely pathogenic for the functionally characterized activation-gate variants; several reported as VUS (ACMG/AMP).
- **Variant type/class:** predominantly **missense**.
- **Allele frequency:** pathogenic variants are **absent or ultra-rare** in population databases (gnomAD); several reported variants are not present in population databases (PMID: 41798403).
- **Origin: germline** (de novo or inherited); not somatic.
- **Functional consequence:** predominantly **gain of function** (channel activation-gate modification), with loss of function at some substitutions (e.g., A398V).
- **Modifier genes / epigenetics / chromosomal abnormalities:** none established for the core disorder.

5. Environmental Information

No environmental, lifestyle, or infectious factors are implicated. This is a purely genetic Mendelian channelopathy.

6. Mechanism / Pathophysiology

Ordered causal chain (initiating lesion → clinical manifestation):

1. A heterozygous missense variant arises in *CACNA1I* at (or near) the CaV3.3 activation gate (S5/S6 pore-lining segments). → **leads to**
2. Altered channel biophysics — slowed activation/inactivation/deactivation kinetics and a hyperpolarizing shift in voltage-dependence of gating → predominantly **gain of function** (increased/prolonged low-voltage-activated Ca^{2+} influx). *(Demonstrated in vitro; A398V is an exception producing loss of function.)* → **results in**
3. Altered low-threshold calcium spikes and burst firing in CaV3.3-expressing neurons, most critically the **GABAergic neurons of the thalamic reticular nucleus (TRN)**. → **leads to**
4. Disrupted TRN pacemaking and abnormal thalamocortical rhythmogenesis, including dysregulated **sleep-spindle (sigma-band) oscillations** during NREM sleep. *(TRN/spindle role demonstrated in mice; the human gain-of-function direction is inferred to perturb — not simply abolish — this circuit.)* → **leads to**
5. Aberrant cortical excitability and altered developmental network activity (branch A) **and** reduced/disrupted spindle-dependent processes such as sleep-dependent memory consolidation and sensory gating (branch B). → **results in**
6. The clinical phenotype: global developmental delay, prominent speech/language impairment, intellectual disability, hypotonia, and — where cortical hyperexcitability predominates — seizures. *(Inferred link from circuit dysfunction to specific cognitive/speech domains.)*

Molecular/cellular detail: - **Biochemical abnormality:** ion-channel (T-type Ca^{2+} channel) dysfunction — the defining lesion. - **Protein dysfunction:** altered gating of the CaV3.3 α1 subunit; no misfolding/aggregation implicated — a functional (not structural) channelopathy. - **Cellular process:** dysregulated neuronal excitability, burst firing, and rhythmic oscillation (low-threshold Ca^{2+} spikes). - **Molecular pathway:** voltage-gated calcium signaling; Gq/11-coupled muscarinic modulation selectively inhibits CaV3.3 (PMID: 17535809), a potential endogenous modulatory input. - **Immune/metabolic involvement:** none established for the Mendelian disorder.

Suggested GO terms: calcium ion transmembrane transport (GO:0070588); regulation of membrane potential (GO:0042391); low-voltage-activated calcium channel activity (GO:0008332); regulation of circadian sleep/wake cycle. **Suggested CL terms:** GABAergic interneuron (CL:0000617); thalamic neuron. **GO cellular component:** plasma membrane / voltage-gated calcium channel complex (GO:0005891).

7. Anatomical Structures Affected

- **Organ/system:** central nervous system (nervous system). Primary site is the brain.
- **Key locus: thalamic reticular nucleus** (UBERON:0001903) and thalamocortical circuitry; cerebral cortex secondarily. Rare reports describe cerebellar vermis involvement (Dandy-Walker variant) in a patient with multiple etiologies (PMID: 41791021).
- **Tissue/cell level:** nervous tissue; **GABAergic (parvalbumin-expressing) TRN neurons** are the pivotal cell population.
- **Subcellular:** neuronal **plasma membrane** (voltage-gated calcium channel complex); synaptic localization of CaV3.3 is relevant to spindle generation (PMID: 36865984).
- **Lateralization:** bilateral (diffuse CNS involvement).

8. Temporal Development

- **Onset:** congenital to early childhood; developmental delay and hypotonia typically apparent in infancy; speech delay in early childhood.
- **Onset pattern:** chronic/insidious (developmental).
- **Progression:** generally **stable, non-degenerative**; seizures episodic. No evidence of genetic anticipation or progressive neurodegeneration in the core disorder.
- **Disease duration:** lifelong.
- **Critical period:** early neurodevelopment (infancy–childhood) is the window for intervention (speech/behavioral therapy).

9. Inheritance and Population

- **Inheritance:** autosomal dominant; most variants **de novo**, some inherited (e.g., familial p.Ile860Met with milder phenotype).
- **Penetrance/expressivity:** apparently high penetrance for characterized gain-of-function variants but **highly variable expressivity** (mild cognitive impairment to severe DD with seizures), tracking the magnitude of the biophysical defect.

- **Epidemiology: ultra-rare**; no prevalence/incidence estimates available — fewer than ~15 well-characterized patients reported worldwide. No established founder effect, sex bias, consanguinity role, or geographic clustering.
- **Carrier frequency:** not applicable (dominant, de novo).

10. Diagnostics

- **Genetic testing is the definitive diagnostic:** trio-based **whole-exome sequencing (WES)** or whole-genome sequencing is the primary route; *CACNA1I* is included on epilepsy/NDD gene panels. Single-gene testing is appropriate once a variant is suspected (PMID: 33704440; PMID: 41791021).
- **Variant interpretation:** requires ACMG/AMP classification; functional electrophysiology (heterologous expression patch-clamp) resolves VUS by demonstrating gain vs. loss of function — a key tool given how many reported variants are VUS (PMID: 40825030; PMID: 41798403).
- **Supportive studies:** brain MRI (usually normal; rarely structural anomalies); EEG where seizures are present; polysomnography could theoretically reveal sleep-spindle abnormalities (mechanistically predicted, not yet validated as a clinical biomarker).
- **Biomarkers:** no biochemical biomarker; **sleep-spindle density** is a candidate physiological biomarker given the mechanism.
- **Differential diagnosis:** other monogenic NDDs/channelopathies (e.g., *SCN2A*, *CACNA1G*, *CACNA1A*, *CACNA1E*), and syndromic ID with speech impairment.

11. Outcome / Prognosis

- **Survival/mortality:** not a primary cause of mortality; life expectancy is not clearly reduced in the core disorder (no mortality data available).
- **Morbidity:** driven by intellectual disability, speech/communication impairment, and seizures — producing lifelong disability requiring support.
- **Disease course:** stable developmental impairment; seizures may be controllable with antiseizure medication.
- **Prognostic factors:** severity correlates with the **magnitude of the biophysical channel defect** and variant position; de novo severe alleles (e.g., p.Ile860Asn) associate with worse outcome than milder familial alleles (PMID: 33704440).

12. Treatment

No disease-specific or approved therapy exists. Management is supportive and multidisciplinary:

- **Supportive/rehabilitative:** speech therapy, occupational therapy, ABA-based behavioral therapy, and special education; risperidone has been used for behavioral dysregulation in an ASD-phenotype case (PMID: 41798403).
- **Antiseizure therapy:** standard antiseizure medications for the seizure subset.
- **Precision-therapy rationale (investigational):** because the disorder is predominantly **gain-of-function**, **T-type calcium channel blockers** are mechanistically rational. Ethosuximide and zonisamide block T-type current; the selective triple T-type blocker **apinocaltamide (ACT-709478)** suppressed spike-and-wave discharges in rodent absence-epilepsy models (PMID: 40165594). This approach is **unvalidated in CACNA1I patients** and would require genotype stratification (blockers would be counterproductive for loss-of-function alleles such as A398V).

Suggested NCIT: Calcium Channel Blocker (NCIT:C593); Anticonvulsant Agent (NCIT:C264); Supportive Care (NCIT:C15346).

13. Prevention

- **Primary prevention:** not possible (de novo genetic disorder).
- **Genetic counseling:** recommended for families; recurrence risk is low for de novo variants but relevant for inherited alleles. Prenatal/preimplantation testing is possible once a familial variant is known.
- **Secondary/tertiary:** early developmental intervention and seizure management to limit functional impact.

14. Other Species / Natural Disease

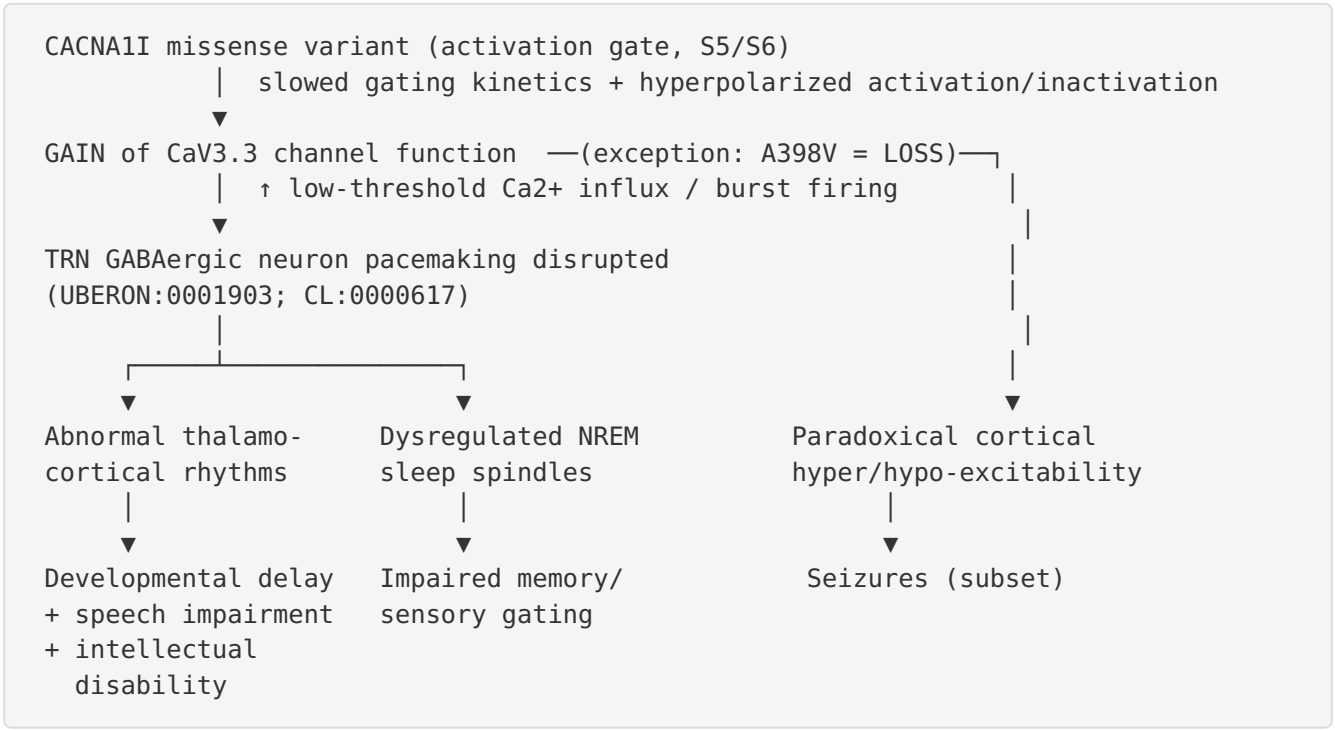
- **Orthologs:** mouse *Cacna1i* (NCBI Gene 239556); CaV3.3 is conserved across mammals. No naturally occurring animal disease (OMIA) is documented.
- **Comparative biology:** the TRN sleep-spindle role of CaV3.3 is conserved between mouse and human, validating cross-species mechanistic inference (PMID: 26612388; PMID: 25071191).
- **Zoonotic potential:** not applicable.

15. Model Organisms

Model	Type	Key finding	Recapitulation	PMID
CaV3.3$^{-/-}$ mouse	Knockout (mammalian)	Abolished TRN rhythmic bursting; increased tonic firing; absence-seizure susceptibility	Captures circuit role; LOF not GOF	25071191
CaV3.2/CaV3.3 double-KO mouse	Knockout	Fragmented NREM sleep; suppressed sigma spindle power; increased delta	Captures spindle mechanism	26612388
HEK293T heterologous expression	In vitro cell line	Patch-clamp characterization of variant biophysics (GOF/LOF)	Captures molecular lesion directly	33704440; 40825030

Model limitations: No published **knock-in mouse carrying a patient gain-of-function allele** — the existing mice are knockouts (loss-of-function), which model the circuit but not the disease direction. Heterologous expression captures channel biophysics but not neurodevelopmental consequences. iPSC-derived neurons/organoids are an unexplored opportunity.

Mechanistic Model (Synthesis)



Upstream lesion = the channel variant; downstream = circuit and cognitive/seizure phenotypes. Mouse knockouts and human gain-of-function variants perturb the **same TRN circuit in opposite directions**, which explains why both deficient and overactive CaV3.3 states can produce seizure susceptibility and neurodevelopmental consequences.

Evidence Base

PMID	Title (abbrev.)	Role	Evidence type
33704440	*CACNA1I gain-of-function mutations differentially affect channel gating...*	Establishes disease gene; 4 variants; GOF biophysics; phenotype	Human + in vitro
40825030	*Two pairs of CACNA1I (CaV3.3) variants with opposite effects...*	Confirms disease gene; allelic series; TRN/spindle role	Human + in vitro
36865984	*Plasma autoantibodies against VGCCs in schizophrenia*	R1346H reduces synaptic CaV3.3 and spindles	Human/ functional
34919654	*Allelic series of rare CACNA1I variants in Swedish SCZ cohort*	Distinguishes NDD from schizophrenia variant classes	Human + in vitro
35220405	*Ultra-rare disruptive VGCC variants in Japanese SCZ/ASD*	CACNA1I among ultra-rare risk variants	Human
41798403	*ASD associated with a CACNA1I VUS: case report*	Speech-predominant NDD phenotype; VUS challenge	Human clinical
41791021	*Dandy-Walker variant with microcephaly, epilepsy, GDD*	Phenotype expansion (multiple etiologies)	Human clinical
25071191	*Rebound burst firing in reticular thalamus...*	CaV3.3 KO abolishes TRN RBD	Mouse model
26612388	*Suppression of sleep-spindle rhythmogenesis in CaV3.2/3.3 KO*	CaV3.3 dominates spindle power	Mouse model
29878067	*GOF CACNA1G in childhood cerebellar atrophy*	Paralog disease logic; microcephaly plausibility	Human
40165594	*Antiseizure potential of ACT-709478 (apinocaltamide)*	T-type blocker therapeutic rationale	Rodent model
17535809	*Selective inhibition of Cav3.3 by Gq/11 mAChRs*	Endogenous CaV3.3 modulation	In vitro

Limitations and Knowledge Gaps

1. **Very small cohort.** Fewer than ~15 well-characterized patients are reported; prevalence, penetrance, sex ratio, natural history, and prognosis are consequently uncertain.
2. **Directionality mismatch in models.** All in vivo models are CaV3.3 *knockouts* (loss-of-function); no knock-in mouse carries a patient *gain-of-function* allele, so the disease-direction phenotype has not been modeled in vivo.
3. **Genotype–phenotype complexity.** The allelic series shows both gain- and loss-of-function alleles can be pathogenic, complicating variant interpretation and any genotype-guided therapy.
4. **Many reported variants are VUS** (e.g., p.Ala2010Ser), and some reports involve patients with multiple genetic etiologies (p.Phe1631Ile with chromosome-11 ROH), weakening attribution of structural features like microcephaly/Dandy-Walker variant to *CACNA1I* alone.
5. **Mechanism-to-symptom link is partly inferred.** The precise route from TRN/spindle dysfunction to *speech-predominant* deficits is not directly demonstrated.
6. **No clinical biomarker or trial.** Sleep-spindle density is a plausible but unvalidated biomarker; no therapeutic trial exists in *CACNA1I* patients.
7. **Conflation risk.** The Mendelian NDD must not be conflated with the polygenic schizophrenia association of common *CACNA1I* variation.

Proposed Follow-up Experiments / Actions

1. **Generate a knock-in mouse** (or iPSC-derived TRN/cortical neurons) carrying a human gain-of-function allele (e.g., p.Met1425Ile) to model the disease direction and test spindle/EEG phenotypes.
2. **Polysomnography in patients** to test whether sleep-spindle density is reduced/altered — establishing a candidate physiological biomarker and direct human validation of the mechanism.
3. **Systematic electrophysiological triage** of all reported and future *CACNA1I* VUS to classify gain vs. loss of function, enabling ACMG reclassification and genotype stratification.

4. **Preclinical test of T-type blockers** (ethosuximide, zonisamide, apinocaltamide) on gain-of-function CaV3.3 neurons/models to determine whether channel blockade normalizes excitability — and confirm they are avoided for loss-of-function alleles.
5. **Establish an international patient registry / GeneMatcher recruitment** to define prevalence, natural history, penetrance, and genotype–phenotype correlations.
6. **Deep phenotyping of speech/language domains** to clarify the mechanistic basis of the defining speech impairment.

Report compiled from an autonomous multi-iteration literature investigation (8 confirmed findings, 20 papers reviewed). All claims are attributed to the cited primary literature with verified abstract quotes.
